Appendix

Seasonal Dynamism or Model Realism: What Drives Better Predictions of Landscape Connectivity?

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Keywords: African wild dog, Connectivity, Dispersal, Dynamism, Individual-based, Lycaon pictus, Okavango Delta, Seasonality, Integrated Step-Selection Functions, Mechanistic Models

Contents

A.1 Seasonality

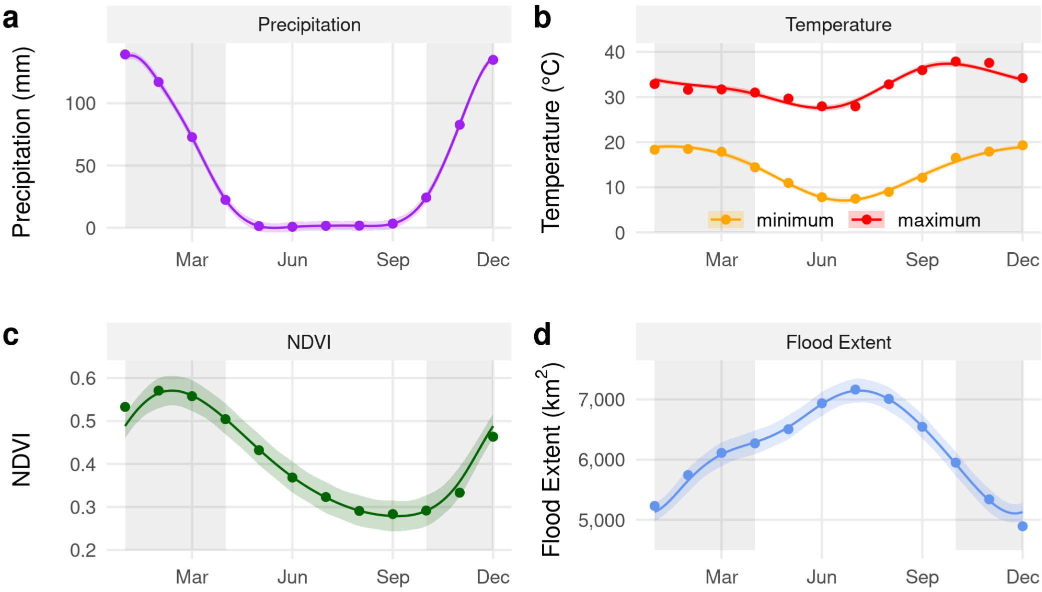

Figure S1: Illustration of how some of the covariates considered in this study vary across seasons. The wet season spans mid-October to mid-April (shaded in gray), the remainder is considered as dry season. Data for the graphs were obtained from (a) JAXA GSMaP (Kubota et al., 2020), (b) ERA5 (Muñoz-Sabater et al., 2021), (c) MODIS MOD13Q1 (Didan, 2015), and (d) remote sensed MCD43A4 satellite images (C. Schaaf, 2015) for the years 2011 to 2022. Smoothing curves were fitted using GAMs as implemented in the mgcv R-package (Wood, 2011). Dots show observed monthly mean values, while ribbons indicate pointwise 95% confidence intervals around the GAM-estimated means.

A.2 GPS Sampling

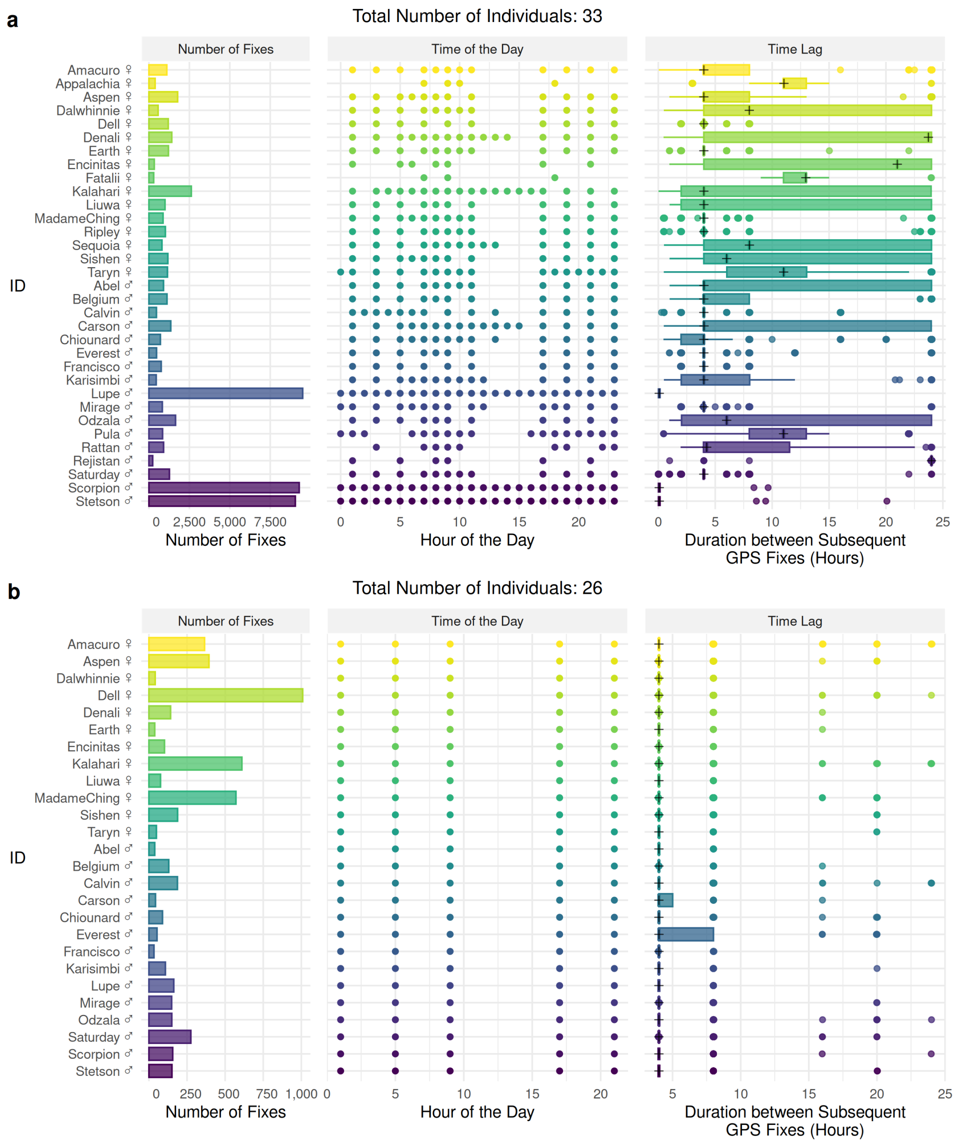

Figure S2: (a) Summary of raw GPS data for all collared individuals, including both resident and dispersing periods. The left panel shows the total number of GPS fixes collected per individual (horizontal bars). The middle panel displays the timing of fixes across the day. The right panel shows the distribution of time lags between successive GPS fixes for each individual, summarized as horizontal boxplots; crosses indicate median fix intervals. (b) Equivalent summaries for the cleaned and standardized dataset, restricted to dispersing individuals and resampled to a consistent fix rate of four hours (and 8 hours during daytime). Because of the standardization, the number of individuals remaining for further analyses dropped from 33 to 26.

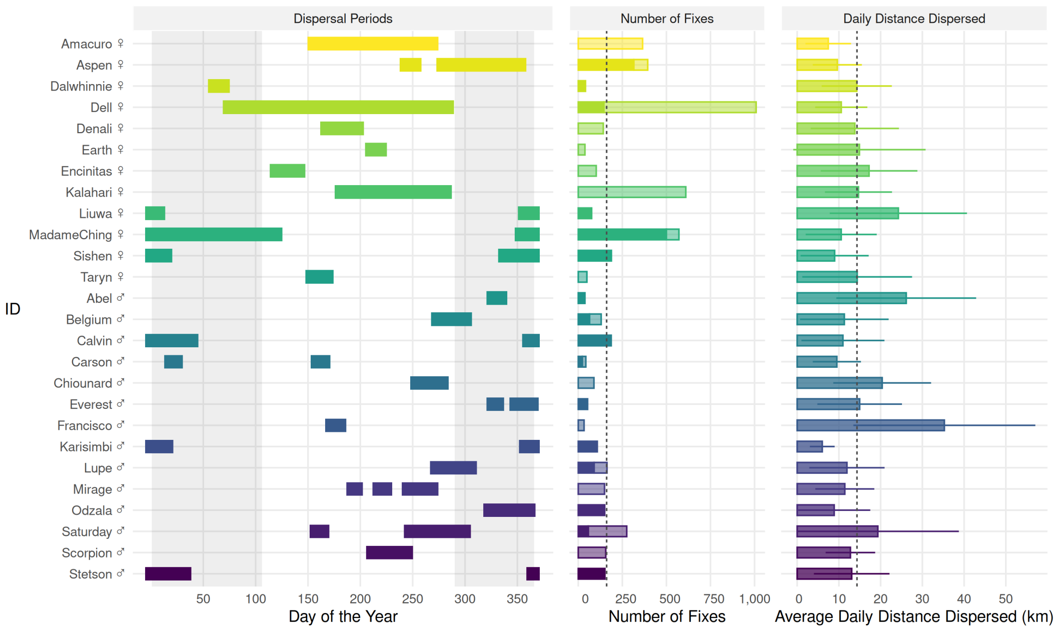

Figure S3: Summary of dispersal-phase GPS data for every collared disperser. The left panel shows the timing and duration of dispersal periods for each individual, plotted by day of year. Some individuals exhibited multiple distinct dispersal events. The dark shaded background denotes the wet season. The middle panel shows the total number of GPS fixes collected per individual, partitioned by season, with dark bars representing fixes collected during the wet season and lighter bars representing fixes collected during the dry season. The right panel shows the mean cumulative (24-h) distance moved by each dispersing individual, with horizontal bars indicating the associated standard deviation.

A.3 Covariates

A.3.1 Landscape Characteristics

We used data from the MODIS Vegetation Continuous Fields dataset (MOD44B V061; DiMiceli et al., 2022) to represent different vegetation types across the study area. The MOD44B dataset comprises three continuous layers, depicting the percentage cover of woodland, shrubs/grassland, and bareland, respectively. The three layers added up to 100%, so we dropped bareland from further analysis, thus preventing perfect multi-collinearity. The MOD44B product is updated on day 65 of each year, so we used the R-package RGISTools (Pérez-Goya et al., 2020) to download yearly updated layers, each at a resolution of 250m x 250m. We also obtained information on the NDVI through the MODIS MOD13Q1 dataset (Didan, 2015), which also has a resolution of 250m x 250m. This product is updated every 16 days, and we accessed the respective data through Google Earth Engine (Gorelick et al., 2017) using the R-package rgee (Aybar et al., 2024). To depict large, permanent waterbodies, we employed the Globeland30 dataset, from which we only retained the land-cover class water, while setting all other categories to dryland. Similarly, we used the MERIT Hydro dataset to obtain information on permanent rivers (Yamazaki et al., 2019). To dynamically render large waterbodies, particularly the floodwaters of the Okavango Delta, we prepared weekly updated floodmaps using remote sensed MODIS MCD43A4 satellite images. The underlying floodmapping algorithm is described in detail by Wolski et al. (2017) and Hofmann et al. (2021) and is implemented in the floodmapr package (available on GitHub; https://github.com/DavidDHofmann/floodmapr). We combined the water (static), river (static), and flood layers (dynamic) into a single stack representing water bodies at a resolution of 500m x 500m. Finally, we employed remote sensing to detect small, ephemeral water bodies (i.e., pans) using a custom random-forest classifier applied to Sentinel 2 satellite imagery (ESA, 2025; details of the classifier in Appendix A2). Sentinel 2 has a resolution of 10m and is therefore particularly useful for obtaining information on small landscape features. Even though Sentinel 2 satellite imagery is updated every five days, cloud cover often prohibited the computation of a “pan-map”. Consequently, we settled for monthly updated composite images, which effectively alleviated problems due to cloud cover. In summary, we produced one stack of layers representing major water bodies, and another stack of layers representing ephemeral water bodies (pans). For both, we also computed corresponding “distance-to-feature” raster layers, indicating the distance to major and ephemeral water bodies, respectively.

A.3.2 Climate Descriptors

We obtained hourly updated spatial layers on 2m above-ground temperature from the ERA5 Land dataset (Muñoz-Sabater et al., 2021) and hourly updated precipitation estimates from the Global Satellite Mapping of Precipitation dataset (Kubota et al., 2020). Both datasets were accessed and downloaded through Google Earth Engine (Gorelick et al., 2017) using the rgee package (Aybar et al., 2024) and had a resolution of 10km x 10km. To match the one-hourly temperature and precipitation estimates with the four-hourly GPS data on dispersing wild dogs, we computed average precipitation and temperature values over four hourly periods that matched the GPS collection schedule.

A.3.3 Anthropogenic Features

We combined information on human density, agricultural activities, and roads into a single proxy, which we generically termed human influence. We sourced information on human density from Facebook's high resolution human density dataset (Tiecke et al., 2017), which we downloaded from the humdata website (www.data.humdata.org). We obtained information on the presence of agricultural fields from the Globeland30 (Chen et al., 2015) and Cropland (Xiong et al., 2017) datasets. We downloaded shapefiles comprising tar and main gravel roads from OpenStreetMap (OpenStreetMap contributors, 2017). Finally, we merged all anthropogenic features into a single layer that had a resolution of 250m x 250m (details are provided by Hofmann et al., 2021.

A.3.4 Light Intensity

We computed light intensity using the `suncalc` and `moonlit` R-packages (Thieurmel & Elmarhraoui, 2022; Śmielak, 2023) for the central coordinates of our study area (lon = of 23.5, lat = -19.0) at a 5-minute temporal resolution. The set of light statistics comprised a binary variable separating day and night (i.e., sun ¡ 18 °below the horizon) and a continuous estimate of moonlight intensity, relative to the maximum moon illumination Figure S4. Based on those covariates, we generated a binary covariate separating bright from dark conditions. Bright conditions encompassed all daytime hours and those nighttime periods during which the moon was present in the sky and illuminated by approximately one-fourth or more; conversely, dark conditions included nighttime periods during which the moon was absent from the sky or present and only minimally illuminated. To match the temporal resolution of our GPS data during dispersal, we calculated four hourly moonlight summaries Figure S5. Specifically, we computed if a four-hourly interval fell into nighttime and the average amount of moon-illumination during those hours.

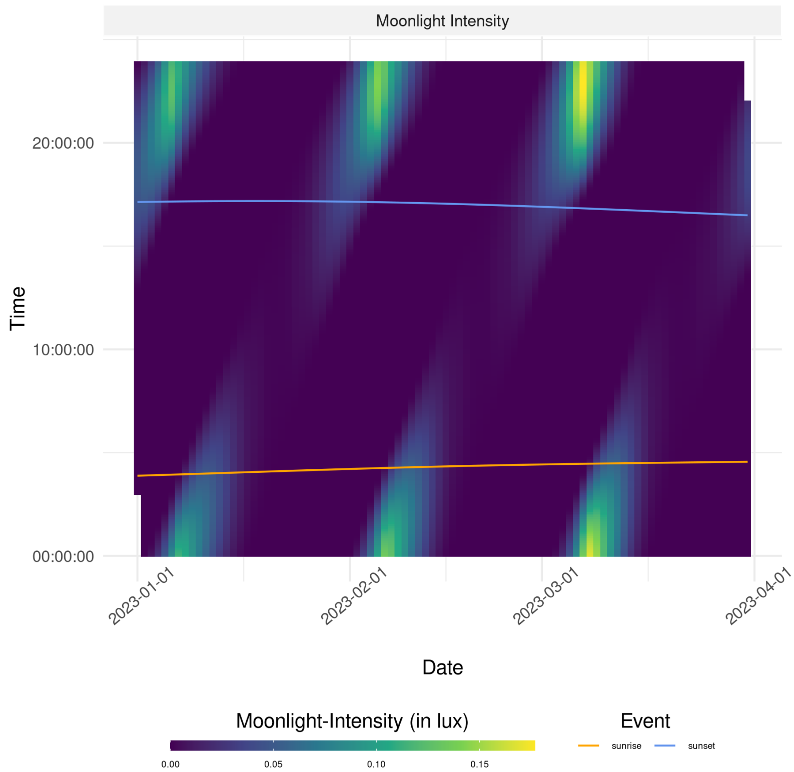

Figure S4: Moonlight intensity (illumination) as estimated from the `moonlit` package (Śmielak, 2023) in `R` calculated for a couple of example timestamps in 2023.

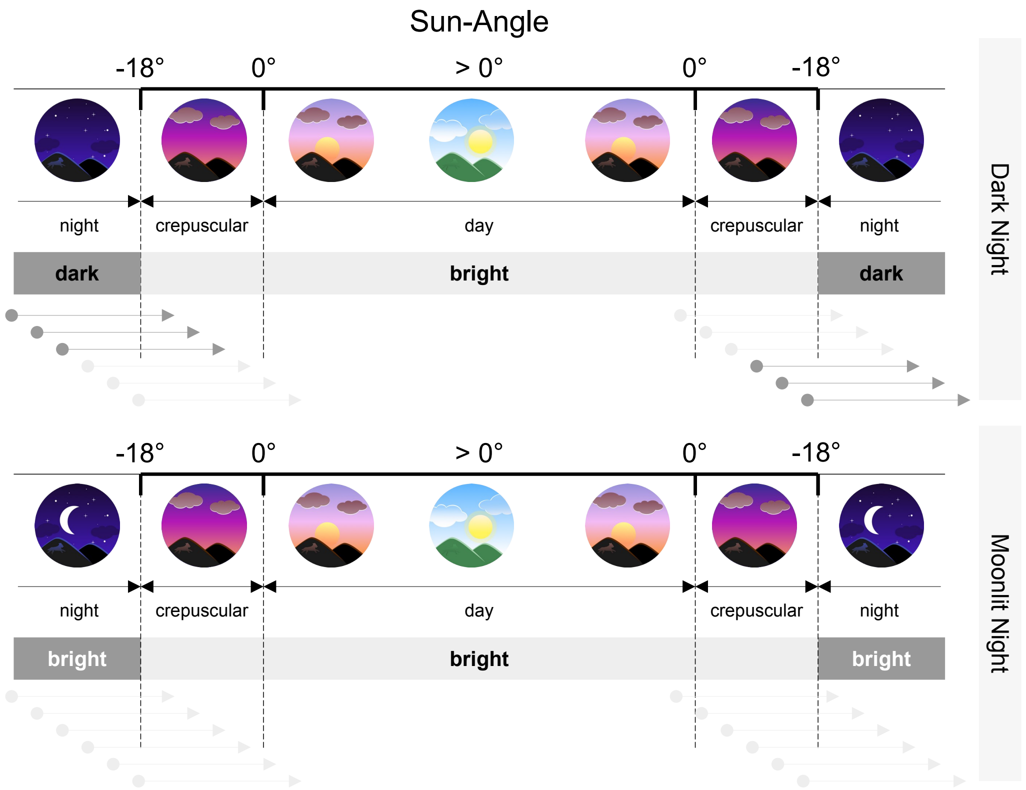

Figure S5: Schematic illustration of how we categorized steps into *dark* and *bright* steps. First, we used the `suncalc` and `moonlit` packages to obtain estimates of the sun-angle and moon illumination at 5 minute intervals for each date of interest. Whenever the sun angle was $\leq$ -18°, we deemed the respective period to be at night. A night was either bright (moon illumination $>$ 0.02 of maximum illumination) or dark (moon illumination $\leq$ 0.02 of maximum illumination). Any other period was considered to provide enough illumination for wild dogs to move and therefore considered as bright. Since each step covered a time-span of approximately four hours, we defined a step as bright if at least 25% (i.e., one hour) of it occurred during a bright period. The gray arrows represent some example steps that are categorized as bright (light gray) or dark (dark gray). All icons are original artwork created by the authors in Inkscape.

A.4 Pan Mapping

A.4.1 Satellite Product Overview

We dynamically mapped the spatial distribution of ephemeral water bodies that span only a few meters in diameter (Figure S6) using a custom remote sensing algorithm. These waterbodies are typically referred to as "pans" and mainly inundate during the rainy season, after which the cumulated water slowly evaporates. Previously, we developed an algorithm to detect large scale floodwaters across the extent of the Okavango Delta. This algorithm made use of MODIS MCD34A4 satellite imagery, which is a 8-day composite of daily updated MODIS satellite data. The benefit of MODIS' high temporal resolution is that monthly composites are almost guaranteed to be cloud free, even during Botswana's rainy season (mid October to mid May). However, due to its coarse spatial resolution of 250 meters, MODIS satellite data was unsuitable to detect finer-scaled waterbodies on satellite images. In an attempt to overcome this limitation, we evaluated alternative satellite products that provided better spatial resolution than MODIS, while retaining a sufficiently high temporal resolution to render seasonal patterns. Candidate products comprised Landsat 7, Landsat 8, and Sentinel 2 satellite imagery. Data associated with each of these satellites are freely accessible and provide spatial resolutions between 10 and 60 meters, depending on the respective bands. Furthermore, the satellites have revisit-times between 5 and 16 days, thus allowing to generate frequently updated composite images (Table S1).

Table S1: Overview of the spatial and temporal resolutions of the candidate satellite products.

Satellite	Availability	Spatial Resolution (m)*	Temporal Resolution (days)
Landsat 7	1999 – Present	15 – 60	16
Landsat 8	2013 – Present	15 – 100	16
Sentinel 2	2015 – Present	10 – 60	5 – 10†

* *Spatial resolutions of the same product can differ depending on the band.*
† *Sentinel 2's revisitation duration decreased to 5 days after March 2017.*

Our goal was to produce dynamically updated pan-maps that we could overlay with GPS data of dispersing African wild dogs. Since GPS data was collected between 2015 and 2022, this implied that satellite data needed to be available for the same period. At first glance, all satellite products appeared to meet this criteria. However, Landsat 7's scan-line detector had been failing since May 31, 2003, introducing data gaps between adjacent tiles, thus rendering the product virtually unusable for our purposes (but see Storey et al., 2005). Sentinel 2 and Landsat 8, by contrast, appeared as viable alternatives, as both of them spanned the desired period and did not exhibit any device failures.

Figure S6: A pan as it can be found in Botswana right after the rainy season. This particular pan is comparably small at around 10 meters in diameter. Some of these pans dry out quickly, whereas others provide a source of water for the entire dry-season. Here, two spotted hyenas (*Crocuta crocuta*) take a quick bath. Photo Credit: David D. Hofmann.

A.4.2 Evaluation of Landsat 8 and Sentinel 2

A.4.2.1 Training Polygons

With Landsat 8 and Sentinel 2 imagery remaining, we conducted a preliminary investigation to compare the two products and evaluate their suitability for our needs. Specifically, we investigated how well we could remote sense pans using either of the two products, utilizing supervised learning methods. For this, we prepared a set of training polygons, consisting of the land cover classes dryland, water, and wet-pans. Although our primary goal was to detect wet-pans, we included the two other classes to facilitate the categorization of reflectance values into distinct groups. Due to the large study area considered, *in situ* ground-truthing of the training polygons was impossible, and we instead opted for *on-screen selection* of training data. More specifically, we utilized Google Earth to digitize areas that were clearly identifiable as either dryland, water, or wet-pans. At the highest zoom level in Google Earth, these categories are visually easy to tell apart (Figure S7). To ensure a sharp distinction between wet-pans and dryland, we traced several dryland polygons in areas that were seasonally covered by water (Figure S7). Google Earth provides the date of any satellite map, and so we could assign a timestamp to each training-polygon. This was necessary to later match training polygons with Landsat or Sentinel data of the same date. Google Earth's ability to display imagery from different dates furthermore allowed us to generate a training dataset that comprised polygons that belonged to different classes depending on the season. For example, a dryland polygon obtained in the dry season could overlap with a wet-pan polygon obtained during the rainy season. Overall, we generated 268 polygons of varying size and shape (104 on dryland, 56 in water, and 108 in wet-pans) for 3 distinct dates (August 18, 2018, March 25, 2019, and July 24, 2021).

A.4.2.2 Satellite Data Download

We downloaded Landsat 8 imagery through Google Earth Engine using the `rgee` `R`-package (Aybar et al., 2024) and used the associated quality band to mask pixels classified as clouds or shadows. We also computed several normalized difference indices (NDs), as listed in Table S2. To download Sentinel 2 data, we used the `sen2r` `R`-package (Ranghetti et al., 2020) which provides a standardized interface to download and process Sentinel 2 imagery from the European Space Agency's data hub. Sentinel 2 data are available as either top-of-the-atmosphere (TOA, level 1C) or bottom-of-the-atmosphere (BOA, level 2A) reflectance values. In analyses where temporal trends are considered, the use of BOA reflectances is recommended, as differences in reflectance properties due to atmospheric conditions are

Figure S7: Example of a *wet pan* and *dryland* training polygon digitized on Google Earth. The extent of water can easily be gauged at the maximum zoom level in Google Earth. While the dryland polygon in this case covered an area that is seasonally covered by water, we placed other dryland polygons in areas that were never inundated. However, to ensure reliable differentiation between wet and dry pans, we included several dryland polygons located in dry pans.

accounted for (Gilabert et al., 1994; Vermote & Kotchenova, 2008; Chraibi et al., 2022). However, a large portion of the publicly available Sentinel 2 data has not yet been processed from level 1C to level 2A, requiring researchers to apply the corrections themselves. Fortunately, the European Space Agency provides the correction tool `sen2cor` for free (Main-Knorn et al., 2017), and this tool has been integrated into the `R`-package `sen2r` (Ranghetti et al., 2020). We therefore applied the necessary corrections after download for any product that hadn't already been corrected. Furthermore, we used Sentinel 2's "Scene Classification Map", to remove pixels classified as either clouds or shadows. Finally, we computed the same ND-indices as for Landsat 8 (Table S2).

Table S2: We computed normalized difference indices between certain bands, hoping they would improve the land-cover classifiers. Depending on the satellite, we used different bands to compute similar indices. The function to compute a normalized difference between bands b_1 and b_2 is given by $\frac{b1-b2}{b1+b2}$.

	Bands	
Index	Landsat 8	Sentinel 2
NDWI	B3, B5	B3, B8
NDMI	B5, B6	B8, B11
NDSI	B3, B6	B3, B11
BEST	B3, B7	B3, B12

A.4.2.3 Land Cover Classifiers

To train a land-cover classifier, we generated 3'000 random points within training polygons following a stratified equal random sampling scheme (Shetty et al., 2021). That is, we ensured that 1'000 random points were sampled per training category. Stratified random sampling was necessary to ensure that wet-pans, our category of interest, was sampled frequently enough, as it represented a minority class, only making up of (0.27%) the total training area. Stratified equal random sampling has been shown to provide good class-level accuracy for minority classes (Shetty et al., 2021), so we deemed this approach suitable for our purposes. We also kept track of the dates of the underlying polygons from which random points were generated. At each random point, we then extracted reflectance values of Sentinel 2 and Landsat 8 data that temporally aligned with the date assigned to the random point (Figure S8a). For instance, if a random point fell into a polygon that was digitized using a Google Earth image generated on Aug 18, 2018, we extracted values from the Sentinel 2 and Landsat 8 layers that were closest to that date. Finally, we parameterized a Random Forest (RF) classifier using the `R`-package `randomForest` (Cutler & Wiener, 2024) and a Classification and Regression Tree (CART) classifier using the `R`-package `rpart` (Therneau et al., 2024). In both cases, we included all bands, as well as derived ND-indices (Table S2) as explanatory covariates. We visualized the decision trees for the CART classifiers (Figure S8b) and variable importance for the RF models (Figure S8c) to investigate the importance of different bands or indices in separating the three classes. To compute variable importance we used the `R`-package `caret` (Kuhn, 2008).

A.4.2.4 Validation and Comparison

To validate the predictive power of the two classifiers across the two datasets, we employed 5-fold cross-validation. For this, we randomly split the data into 5 groups and repeatedly fitted both classification models using 80% of the data, to then predict land cover categories in the remaining 20%. We generated confusion matrices to contrast true and predicted labels and computed estimates of the classifier's specificity, sensitivity, and overall accuracy. The results from this validation show that both classifiers achieve $\geq 90\%$ accuracy across both satellite products (Figure S9). In fact, the RF classifier resulted in an overall accuracy of 99% on Sentinel 2 data.

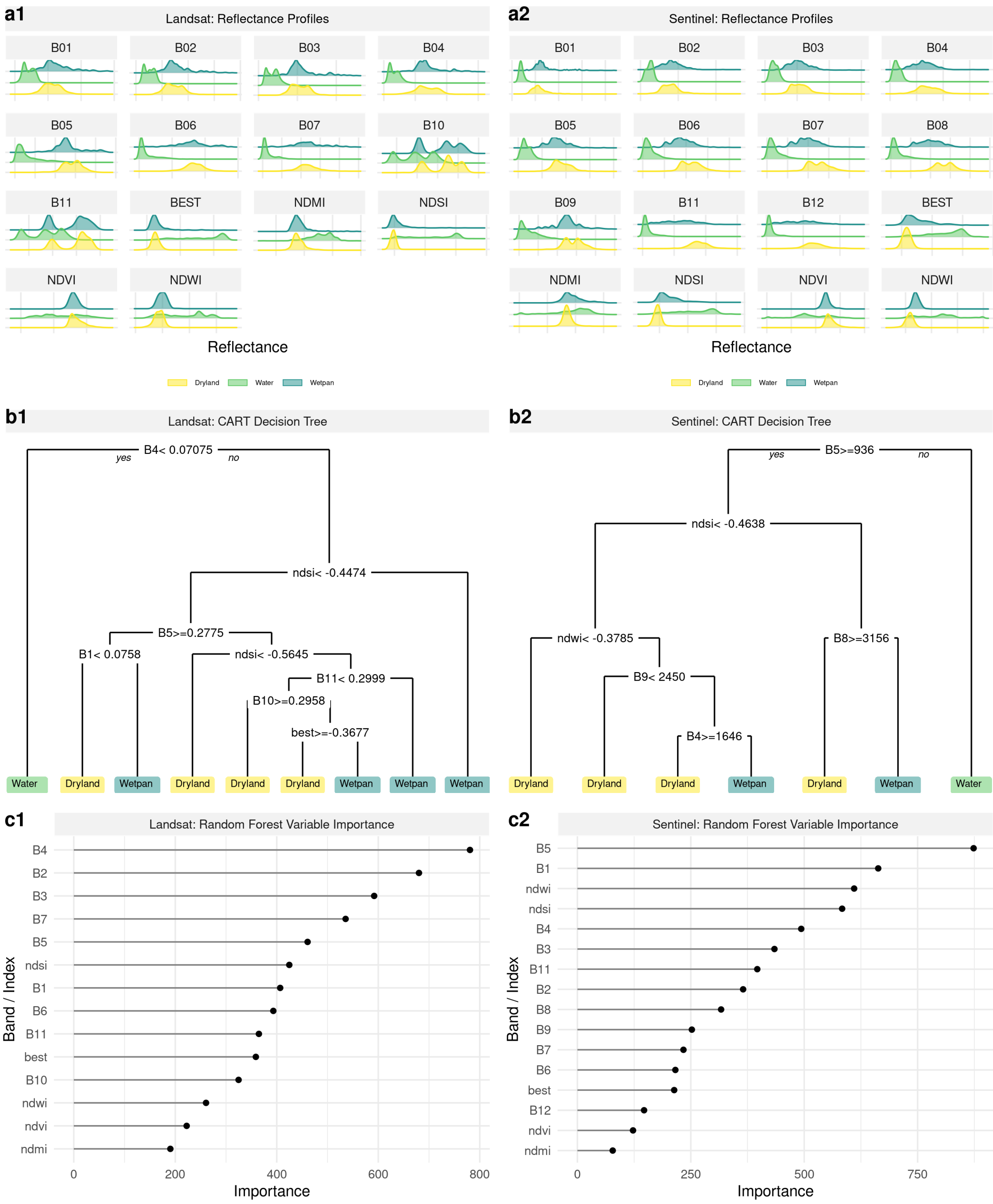

Figure S8: Reflectance properties of the Landsat 8 (a1) and Sentinel 2 (a2) bands and NDs at the extracted training points for each of the three categories (colored). Based on extracted reflectance values we parametrized Classification and Regression Tree models (b1 and b2) as well as Random Forest models (c1 and c2).

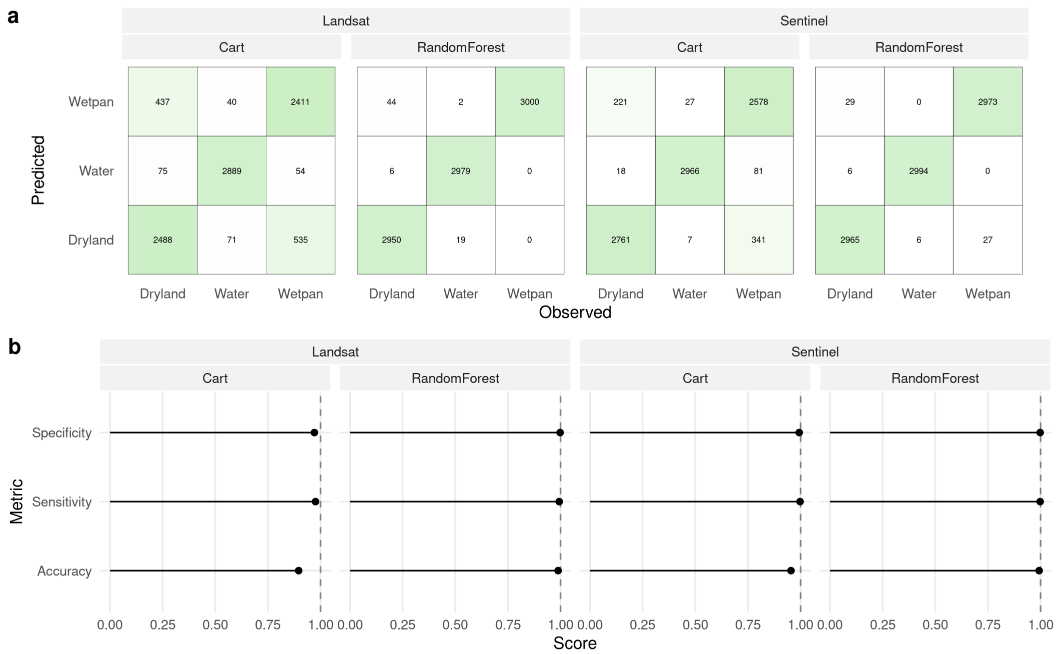

Figure S9: Confusion matrices (a) and derived performance metrics (b) for the CART and RF classifiers for both the Landsat 8 and Sentinel 2 datasets.

A.4.3 Bulk Download

Although both Landsat 8 and Sentinel 2 provided very good results, we considered Sentinel 2 data to be marginally superior, mainly due to Sentinel's higher temporal and spatial resolution. A higher temporal resolution was key to compensate missing data from satellite mages obtained on cloudy days and therefore pivotal in achieving cloud free monthly composites. We therefore decided to download Sentinel 2 data for the period during which we also collected GPS data of dispersing individuals. Instead of downloading all Sentinel 2 tiles overlapping with our study area and matching the study-period, we created a spatio-temporal moving window (Figure S10). This window was updated each month and comprised all GPS locations collected during that month, buffered by a 100 km radius. We then identified Sentinel 2 tiles intersecting with each of the monthly updated moving windows (Figure S11). This resulted in a list of 2,226 tiles that needed to be downloaded, 1,373 of which were already corrected to BOA reflectances, while the remaining 853 tiles still needed to be corrected. The download followed the same procedure and included the same corrections (from TOA to BOA, and cloud masking) as outlined in Appendix A.4.2.2 using the `sen2r` package in `R` (Ranghetti et al., 2020). Upon completion of the download and pre-processing, we applied the trained RF classifer to obtain binary "pan-maps", showing the spatial distribution of ephemeral water bodies. To generate monthly composites, we merged all pan-maps falling into the same month and retained pans if they were detected in at least 50% of the overlapping tiles. Finally, we generated "distance-to-pan" maps using the `distance` function from the `terra` package in `R` (Hijmans et al., 2024).

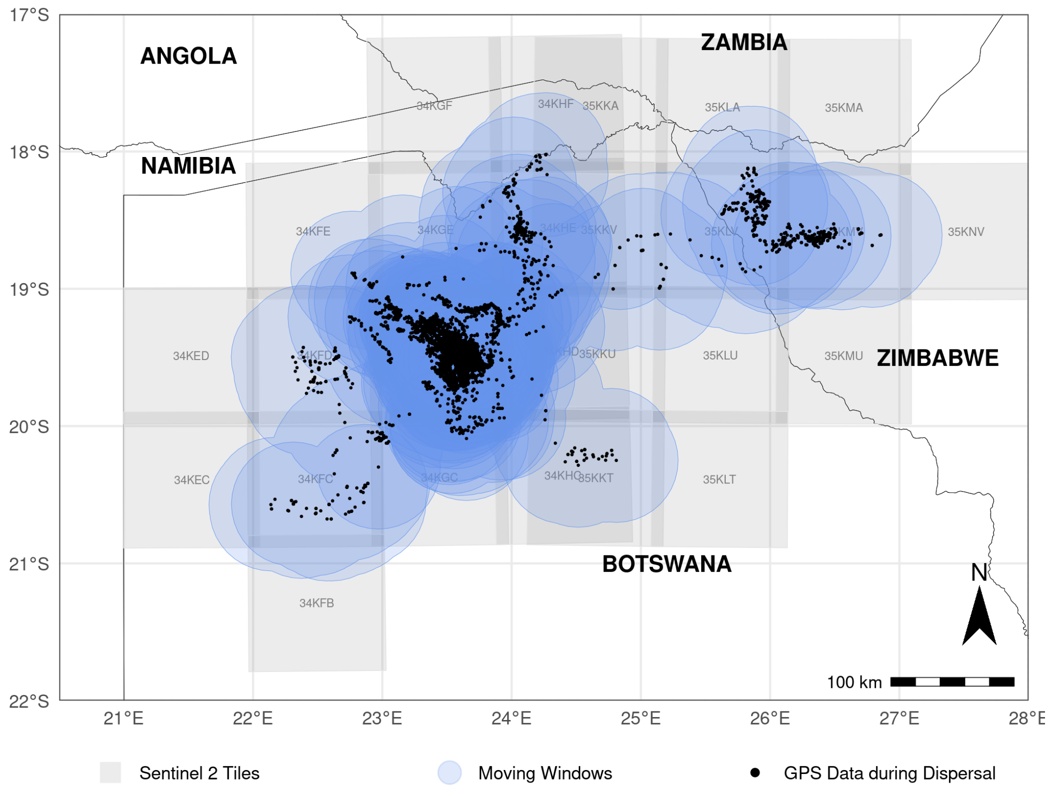

Figure S10: To reduce the number of Sentinel 2 tiles that we needed to download, we generated a monthly updated spatio-temporal moving window that comprised all GPS data of dispersing wild dogs (black dots) during the respective month, buffered by a 100 km radius. Based on the so created moving windows (in blue), we identified all overlapping Sentinel 2 tiles (tiles in gray) that we needed to download each month, which resulted in a total of 2,226 tiles.

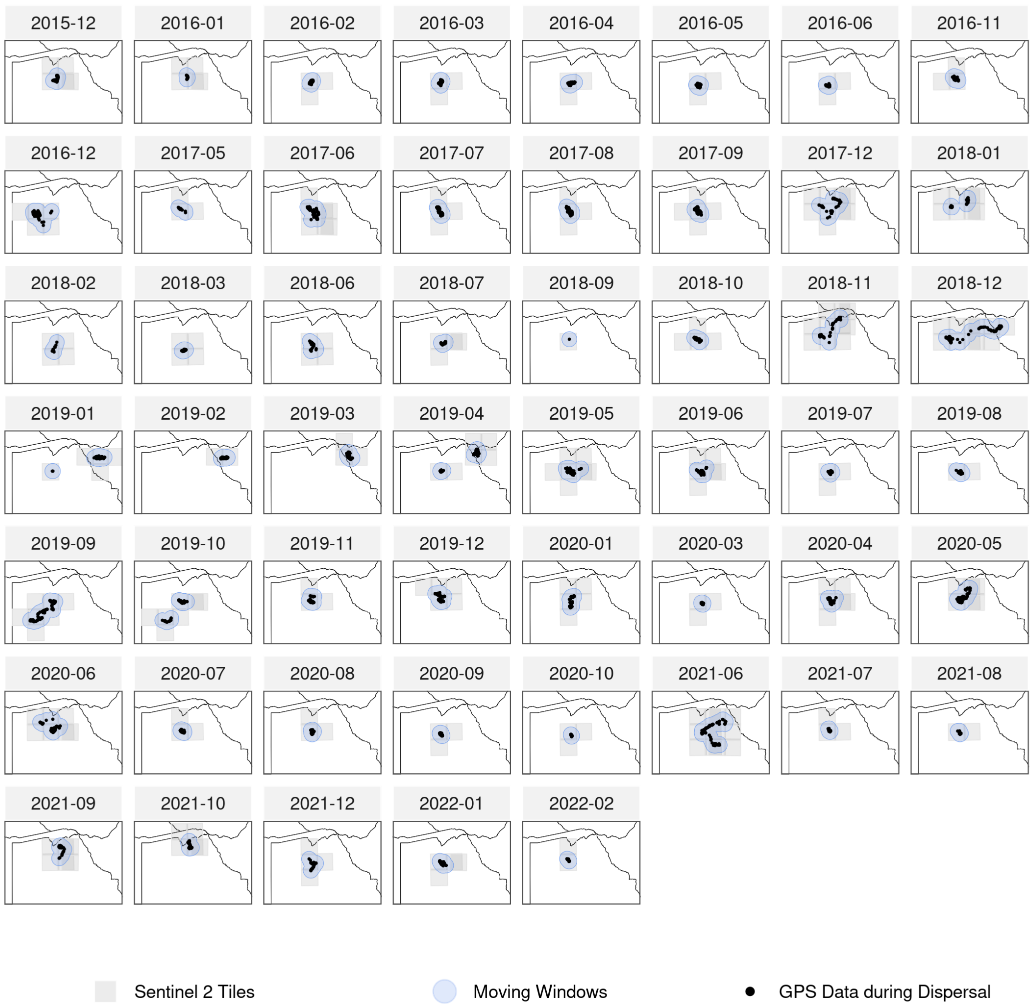

Figure S11: Moving windows and associated tiles facetted by months.

A.5 Movement Model Results

Table S3: Estimates obtained from the **simplistic** (no interactions) integrated step-selection models fitted using **static** covariates. GPS Data was either pooled across seasons (labeled "all") or split into dry and wet season. Only underlined covariates differed between the static and dynamic configurations.

		Fixed Effects			Random Effects
Season	Covariate	Coefficient	p-value	Significance	SD
All	sl	0.058	0.214		0.203
	log(sl)	0.057	0.069	*	0.109
	cos(ta)	0.111	0.000	***	0.053
	Humans	-0.289	0.019	**	0.358
	Trees	-0.279	0.011	**	0.376
	Shrubs	0.139	0.492		0.170
	Water	-0.467	0.018	**	0.224
	DistanceToWater$^{0.5}$	-0.780	0.000	***	0.604
Dry	sl	0.047	0.376		0.170
	log(sl)	0.083	0.004	***	0.043
	cos(ta)	0.120	0.000	***	0.060
	Humans	-0.219	0.110		0.302
	Trees	-0.327	0.031	**	0.450
	Shrubs	-0.091	0.730		0.321
	Water	-0.775	0.001	***	0.003
	DistanceToWater$^{0.5}$	-0.932	0.000	***	0.681
Wet	sl	0.010	0.889		0.257
	log(sl)	0.032	0.497		0.128
	cos(ta)	0.116	0.001	***	0.083
	Humans	-0.271	0.140		0.366
	Trees	-0.059	0.499		0.000
	Shrubs	0.495	0.114		0.007
	Water	0.113	0.711		0.152
	DistanceToWater$^{0.5}$	-0.550	0.014	**	0.619

Table S4: Estimates obtained from the **simplistic** (no interactions) integrated step-selection models fitted using **dynamic** covariates. GPS Data was either pooled across seasons (labeled "all") or split into dry and wet season. Only underlined covariates differed between the static and dynamic configurations.

		Fixed Effects			Random Effects
Season	Covariate	Coefficient	p-value	Significance	SD
All	sl	0.050	0.270		0.199
	log(sl)	0.064	0.045	**	0.110
	cos(ta)	0.112	0.000	***	0.052
	Humans	-0.252	0.036	**	0.340
	<u>Trees</u>	-0.216	0.010	**	0.336
	<u>Shrubs</u>	-0.012	0.909		0.239
	<u>Water</u>	-0.680	0.000	***	0.252
	<u>DistanceToWater</u>$^{0.5}$	-0.235	0.089	*	0.523
Dry	sl	0.035	0.506		0.175
	log(sl)	0.082	0.001	***	0.000
	cos(ta)	0.118	0.000	***	0.057
	Humans	-0.176	0.174		0.271
	<u>Trees</u>	-0.360	0.004	***	0.413
	<u>Shrubs</u>	-0.151	0.298		0.340
	<u>Water</u>	-0.877	0.000	***	0.000
	<u>DistanceToWater</u>$^{0.5}$	-0.309	0.082	*	0.529
Wet	sl	0.012	0.864		0.251
	log(sl)	0.047	0.339		0.137
	cos(ta)	0.123	0.000	***	0.085
	Humans	-0.232	0.203		0.367
	<u>Trees</u>	-0.070	0.341		0.156
	<u>Shrubs</u>	0.253	0.127		0.273
	<u>Water</u>	-0.352	0.058	*	0.194
	<u>DistanceToWater</u>$^{0.5}$	-0.163	0.410		0.558

Table S5: Estimates obtained from the **mechanistic realistic** (with interactions) integrated step-selection models fitted using **static** covariates. GPS Data was either pooled across seasons (labeled "all") or split into dry and wet season. Only underlined covariates differed between the static and dynamic configurations.

		Fixed Effects			Random Effects
Season	Covariate	Coefficient	p-value	Significance	SD
All	sl	-0.014	0.734		0.152
	log(sl)	0.478	0.000	***	0.138
	cos(ta)	0.128	0.000	***	0.060
	sl:Dark	-0.294	0.000	***	-
	log(sl):Dark	-0.911	0.000	***	-
	sl:Temperature	-0.234	0.000	***	-
	log(sl):Temperature	-0.197	0.000	***	-
	Humans	-0.317	0.019	**	0.412
	Trees	-0.241	0.038	**	0.415
	Shrubs	0.173	0.397		0.128
	Water	-0.507	0.012	**	0.253
	DistanceToWater$^{0.5}$	-0.885	0.000	***	0.672
	sl:Trees	0.001	0.974		-
	sl:Shrubs	0.081	0.561		-
	sl:Water	-0.183	0.174		-
	cos(ta):DistanceToWater$^{0.5}$	0.058	0.000	***	-
Dry	sl	-0.057	0.311		0.163
	log(sl)	0.532	0.000	***	0.160
	cos(ta)	0.128	0.000	***	0.055
	sl:Dark	-0.122	0.070	*	-
	log(sl):Dark	-1.036	0.000	***	-
	sl:Temperature	-0.228	0.000	***	-
	log(sl):Temperature	-0.248	0.000	***	-
	Humans	-0.259	0.087	*	0.353
	Trees	-0.276	0.086	*	0.481
	Shrubs	-0.058	0.827		0.238
	Water	-0.846	0.001	***	0.209
	DistanceToWater$^{0.5}$	-1.098	0.000	***	0.796
	sl:Trees	0.050	0.342		-
	sl:Shrubs	-0.046	0.793		-
	sl:Water	-0.343	0.042	**	-
	cos(ta):DistanceToWater$^{0.5}$	0.035	0.107		-
Wet	sl	-0.041	0.568		0.217
	log(sl)	0.404	0.000	***	0.153
	cos(ta)	0.142	0.000	***	0.112
	sl:Dark	-0.894	0.000	***	-
	log(sl):Dark	-0.726	0.000	***	-
	sl:Temperature	-0.251	0.000	***	-
	log(sl):Temperature	-0.140	0.004	***	-
	Humans	-0.265	0.181		0.420
	Trees	-0.034	0.712		0.000
	Shrubs	0.562	0.092	*	0.141
	Water	0.136	0.671		0.001
	DistanceToWater$^{0.5}$	-0.578	0.007	***	0.571
	sl:Trees	-0.074	0.277		-
	sl:Shrubs	0.383	0.101		-
	sl:Water	0.141	0.534		-
	cos(ta):DistanceToWater$^{0.5}$	0.090	0.001	***	-

Table S6: Estimates obtained from the **mechanistic realistic** (no interactions) integrated step-selection models fitted using **dynamic** covariates. GPS Data was either pooled across seasons (labeled "all") or split into dry and wet season. Only underlined covariates differed between the static and dynamic configurations.

		Fixed Effects			Random Effects
Season	Covariate	Coefficient	p-value	Significance	SD
All	sl	-0.172	0.002	***	0.223
	log(sl)	0.435	0.000	***	0.136
	cos(ta)	0.110	0.000	***	0.058
	sl:Dark	-0.305	0.000	***	-
	log(sl):Dark	-0.805	0.000	***	-
	sl:Temperature	-0.187	0.000	***	-
	log(sl):Temperature	-0.083	0.000	***	-
	Humans	-0.278	0.040	**	0.401
	Trees	-0.208	0.016	**	0.347
	Shrubs	-0.018	0.866		0.223
	Water	-0.791	0.000	***	0.176
	DistanceToWater$^{0.5}$	-0.306	0.030	**	0.537
	sl:Trees	-0.028	0.141		-
	sl:Shrubs	0.051	0.462		-
	sl:Water	-0.435	0.000	***	-
	cos(ta):DistanceToWater$^{0.5}$	0.046	0.005	***	-
Dry	sl	-0.267	0.000	***	0.181
	log(sl)	0.454	0.000	***	0.116
	cos(ta)	0.115	0.000	***	0.054
	sl:Dark	-0.124	0.062	*	-
	log(sl):Dark	-0.902	0.000	***	-
	sl:Temperature	-0.180	0.000	***	-
	log(sl):Temperature	-0.087	0.000	***	-
	Humans	-0.220	0.149		0.348
	Trees	-0.353	0.005	***	0.414
	Shrubs	-0.137	0.343		0.319
	Water	-0.957	0.000	***	0.001
	DistanceToWater$^{0.5}$	-0.373	0.036	**	0.526
	sl:Trees	-0.018	0.414		-
	sl:Shrubs	0.009	0.916		-
	sl:Water	-0.345	0.002	***	-
	cos(ta):DistanceToWater$^{0.5}$	0.007	0.731		-
Wet	sl	-0.077	0.401		0.284
	log(sl)	0.427	0.000	***	0.165
	cos(ta)	0.125	0.002	***	0.111
	sl:Dark	-0.907	0.000	***	-
	log(sl):Dark	-0.667	0.000	***	-
	sl:Temperature	-0.238	0.000	***	-
	log(sl):Temperature	-0.086	0.037	**	-
	Humans	-0.203	0.307		0.430
	Trees	-0.059	0.414		0.143
	Shrubs	0.279	0.082	*	0.217
	Water	-0.454	0.009	***	0.001
	DistanceToWater$^{0.5}$	-0.177	0.378		0.571
	sl:Trees	-0.073	0.072	*	-
	sl:Shrubs	0.186	0.124		-
	sl:Water	-0.493	0.002	***	-
	cos(ta):DistanceToWater$^{0.5}$	0.111	0.000	***	-

A.6 Model Sensitivity to Number of Random Steps

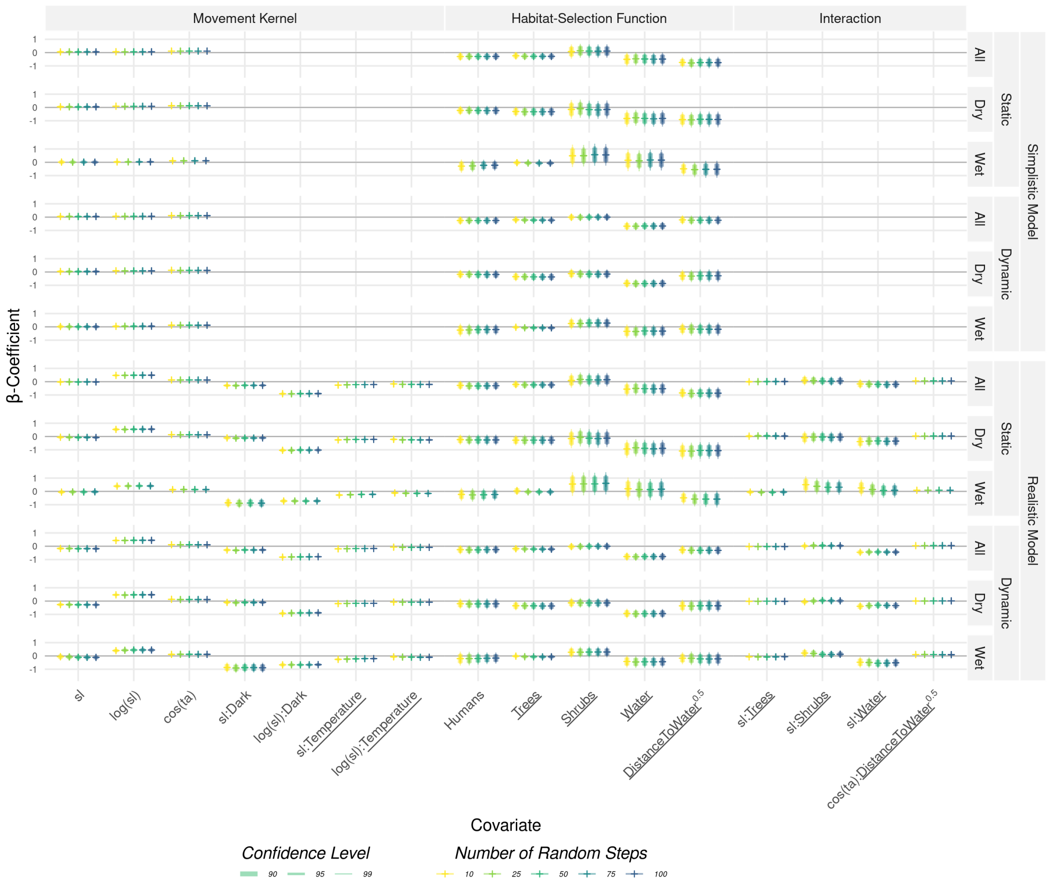

Figure S12: Results from the integrated step-selection models when considering different numbers of random steps. Results are shown for the simplistic and mechanistic realistic model for both all configurations of fitting covariates (static vs dry) and model seasons (all vs. wet + dry).

A.7 Model Interpretation

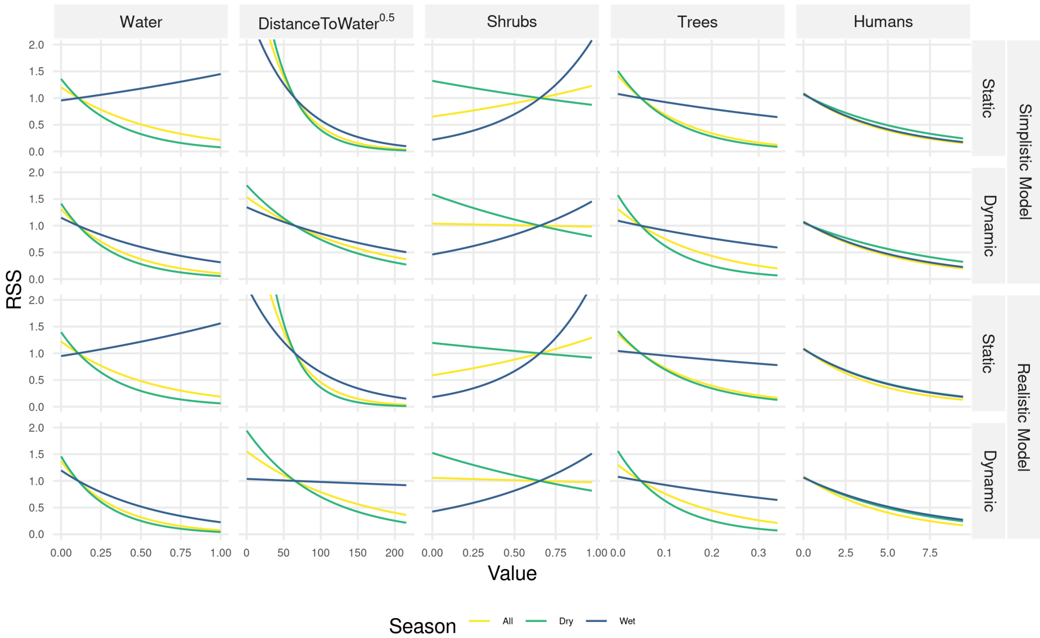

Figure S13: Relative selection scores (RSS) with respect to environmental covariates. RSS scores were computed by comparing a location x_1 with average conditions to a location x_2 with average conditions but the respective covariate varied between its minimum and maximum observed value. Confidence intervals overlapped substantially, hence we omitted them from the figure to simplify with the interpretation of the differences among configurations.

A.8 Spearman's Rank Correlation in Relation to the Number of Random Steps

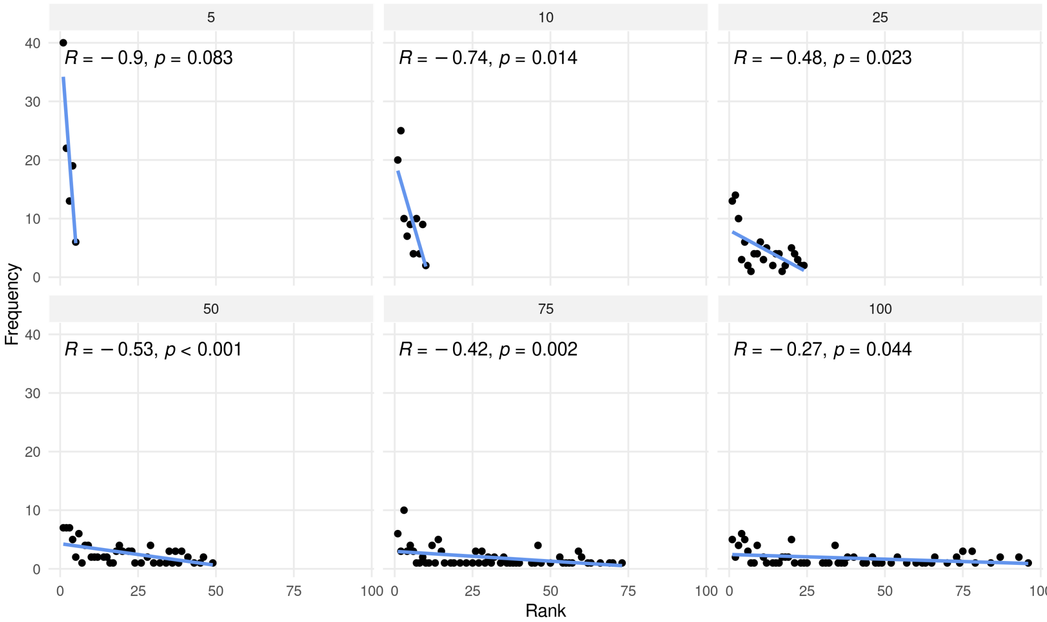

Figure S14: Illustration how the Spearman's rank correlation changes when the number of random steps per stratum is varied from 5 to 100 (facettes). The y-axis shows the frequency at which the observed step was assigned the rank indicated on the x-axis. A low rank indicates that the observed step was assigned a high probability of being chosen. The better the prediction, the more frequently the observed step should be assigned a low rank, thus resulting in negative Spearman's rank-correlation. However, it is obvious that the metric heavily depends on the number of random steps per stratum. If there are only a few random steps, the correlation is more likely to be negative. The show patterns are based on simulated data.

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